

Dhanya Nair, CEng MIMechE MAPM

Chartered Mechanical Engineer | Semiconductor Equipment | Manufacturing Quality | Program Delivery

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Location: Bangalore, India

PERSONAL STATEMENT

Chartered Mechanical Engineer (CEng MIMechE) and APM-qualified engineering professional with 14+ years' experience in mechanical product development and engineering delivery across semiconductor equipment, scientific infrastructure and automotive sectors, including six years at Lam Research. Strong background in end-to-end product development, mechanical design, GD&T, DFM, supplier engagement, prototyping, validation and production release, with experience supporting manufacturing quality, dimensional inspection and structured problem solving as part of engineering delivery. Experienced in cross-functional technical leadership, project planning and supplier management, with experience in engineering data analysis and developing capability in Python, SQL and machine learning. Seeking to combine mechanical engineering, manufacturing and program experience with growing data and digital capabilities to support engineering transformation.

CAREER HISTORY

Daresbury Laboratory, STFC, United Kingdom – Senior Mechanical Engineer (May 2024 – June 2026)

Anode Plane Assembly (APA) for Deep Underground Neutrino Experiment (DUNE)

- Led mechanical design and reliability improvements for production winder equipment, investigating issues including interference, alignment and mechanical failures, and developing design modifications to improve equipment function and robustness.
- Reviewed legacy equipment designs and historical modifications against as-built conditions; resolved discrepancies between drawings and installed equipment, assessed previous design changes for suitability and reliability, and updated engineering documentation to reflect the current configuration.
- Developed the equipment layout and safety concept for installation of additional winding assemblies within a space-constrained production area, including guarded access and a sliding light-curtain support arrangement to enable safe operation and maintenance.
- Designed and developed test fixtures and production-support equipment to validate wire tension; produced manufacturing drawings to BS 8888 with GD&T for machined and fabricated components, and validated fit using rapid prototypes prior to manufacture.
- Worked closely with technicians during live production to investigate mechanical issues and non-conformances, develop practical engineering solutions and minimise disruption to the build programme.
- Developed preventive-maintenance plans for critical winder sub-assemblies, identifying wear components and defining replacement schedules to improve equipment reliability and uptime.
- Led a Six Sigma DMAIC process-improvement project implementing a structured inventory and shipment-tracking workflow for mechanical parts, reducing production downtime by 45% through improved part visibility, availability and reorder control.

- Conducted supplier-site and in-house dimensional and visual inspections of machined and welded structural assemblies, verifying dimensional conformity, fabrication quality and workmanship against engineering drawings and inspection requirements, including visual assessment of weld quality and identification of non-conformances prior to release for manufacture and assembly.
- Helped develop and improve inspection procedures by clarifying inspection steps, acceptance requirements and visual guidance.
- Led supplier and vendor interface activities including RFQs, technical drawing clarifications, material-selection discussions, manufacturing queries and procurement coordination for machined, fabricated and off-the-shelf components.
- Planned multi-year procurement requirements from engineering BOMs in collaboration with the project management team, prioritising long-lead items, storage constraints and available budget. Identified established components suitable for bulk purchasing, achieving cost efficiencies.

Jaguar Land Rover, United Kingdom – Lead Mechanical Engineer (Nov 2022 – May 2024)

Electric Charging System for 1st Electric Modular Longitudinal Architecture Vehicle

- Led mechanical design and development activities for fuel and charge flap systems for JLR's first Electric Modular Longitudinal Architecture vehicle, including CAD design reviews, design optimisation, GD&T and tolerance stack-up analysis to achieve fit, function and integration within tight vehicle packaging constraints.
- Worked with international Tier-1 suppliers in Germany and Poland on injection-moulded and electromechanical components, providing technical direction on design, manufacturability, quality issues and validation requirements through remote collaboration and supplier engagement.
- Supported verification builds and lifecycle testing for fuel and charge flap systems, acting as a technical point of contact to coordinate test requirements, test requests, schedules, parts and materials across engineering, test teams and suppliers. Worked with subject-matter experts to translate validation requirements into test specifications and ensure test readiness and execution.
- Investigated production-quality issues affecting fuel and charge flap gap and flush. Conducted factory-floor dimensional studies across multiple vehicles using defined measurement locations and engineering tolerance requirements, including repeated measurements by different operators to assess measurement variation.
- Analysed measurement data in Minitab to evaluate variation and distribution against engineering requirements, presented findings to engineering stakeholders, and supported root-cause investigation and corrective actions using RCCA and 5 Why.
- Supported verification build and issue resolution through structured troubleshooting and root-cause analysis, implementing short-term containment and longer-term corrective actions. Represented component issues in daily verification-build stand-ups and communicated findings and lessons learned back to the engineering team.
- Worked in an Agile engineering environment using Jira and Kanban, participating in two-week sprint planning, daily stand-ups, design reviews and programme meetings to manage engineering priorities, requirements and cross-functional delivery.

Lam Research, India – Senior Mechanical Engineer (Aug 2015 - Aug 2021)

PECVD and ALD Wafer Fabrication Equipment (Altus Max and Vector Platforms)

- Delivered end-to-end engineering of complex electromechanical subsystems for Lam PECVD and ALD wafer-fabrication equipment, from concept development through detailed design and analysis, supplier engagement, prototype builds, validation and production release.
- Designed sheet metal enclosures, machined components, and plastic parts applying DFM principles, including injection moulding constraints, bending allowances, and tooling feasibility ensuring cost-effective and manufacturable designs.
- Created detailed 3D CAD models and manufacturing drawings with GD&T to ASME standards, defining dimensional and tolerance requirements for machined, sheet-metal and assembled components; performed tolerance stack-up analysis to verify fit, alignment and functional compliance.
- Performed thermal and structural analysis to optimise component temperatures, material selection, and mechanical robustness including heat sink sizing, fan selection, and validation against performance and reliability requirements.
- Ensured all designs met SEMI standards (S2, S22) and applicable electrical safety requirements including creepage and clearance analysis per IEC standards.
- Developed design verification plans (DVP), supported prototype builds and investigated non-conformances using structured problem-solving methods including DFMEA, RCCA, 5 Why and 8D.
- Served as technical owner across concurrent semiconductor-equipment projects, planning engineering activities, managing priorities and technical risks, and driving milestones from concept through prototype build, validation and HVM production release.
- Coordinated cross-functional engineering activities across mechanical, electrical, manufacturing, supply chain, program management, safety and suppliers to resolve design, build and production issues throughout the product lifecycle.
- Managed technical interfaces with international suppliers through remote collaboration and on-site engagement, covering DFM reviews, engineering-drawing clarifications, manufacturing queries, material selection and production-component delivery.
- Resolved high-priority problem reports across multiple deposition-tool platforms, supporting root-cause investigation, engineering drawing updates and design iterations.

UST Global, India –Design Engineer (Aug 2011 - Feb 2015)

Low Voltage Motor Control Center (LVMCC)

- Generated mechanical layout, air flow design, and sealing solutions for LVMCC used in industrial automation applications.
- Performed creepage and clearance analysis to ensure electrical safety.
- Harness creation and drawing flattening for low voltage motor drives.
- Product lifecycle and engineering-data management including CAD file management, part-number creation, BOM maintenance and release control.

EDUCATION

MSc Advanced Mechanical Engineering: Cranfield University, Cranfield, UK (Oct 2021 - Sept 2022)

- **Modules:** Engineering Stress Analysis, Risk and Reliability Engineering, Structural Integrity, Computational Fluid Dynamics for Renewable Energy, Applied Materials and Corrosion, Component Design, Engineering Project Management.
- **Group Design:** Feasibility study of heat pump technology utilisation at Railway corridors
- **Individual Thesis:** Structural integrity assessment of weld cracks in a reactor vessel

BE Mechanical Engineering: Visveswaraya Technological University, India (Oct 2007 - Jun 2011)

- **Modules:** Material Science and Metallurgy, Thermodynamics, Mechanics of Materials, Fluid Mechanics, Design of Machine Elements, Heat and Mass Transfer.
- **Group Design:** Test Bench design for conducting ground test on Hydraulic Tail Rotor Actuator

COMPETENCIES AND SKILLS

- **Manufacturing & Quality:** GD&T, tolerance analysis, CTQ characteristics, dimensional inspection, DFM, statistical quality analysis using Minitab, FMEA, RCCA, 5 Why, 8D, Six Sigma DMAIC
- **Program & Supplier Management:** Cross-functional technical delivery, risk and issue management, supplier management, NPI/HVM production release, Agile, Jira/Kanban, MS Project
- **Data & Digital:** Python, SQL (basic), machine learning fundamentals
- **Engineering Tools & Systems:** Creo, SolidWorks, CATIA, Abaqus, ANSYS, Teamcenter, Windchill, SAP
- **Standards & Regulatory Knowledge:** SEMI (S2, S22), ANSI, ASTM, BS 8888, ISO 9001:2015 (Quality Management Systems); familiarity with IEC 61439, IEC 60204-1

PROFESSIONAL STATUS

- Chartered Engineer (CEng), Institution of Mechanical Engineers (IMechE), UK
- Member, Association for Project Management (MAPM), UK
- APM Project Management Qualification (PMQ), UK
- Six Sigma Yellow Belt certified